Вариант № 4

1. $tgx \cdot y'' - y' + \frac{1}{\sin x} = 0$
2. $y'\sin x - y\cos x = 1$ $\qquad y\left(\frac{\pi}{2}\right) = 0$
3. $y'' - 3y' = 3x + x^2$
4. $y' - y = xy^2$
5. $y^3 y'' = -1$
6. $6xdx - 6ydy = 3x^2 ydy - 2xy^2 dx$
7. $xyy' = y^2 - xy - x^2$
8. $2y'' - 13y' - 7y = \sin 2x$
9. $y'' + y = \frac{1}{\sin x}$

Вариант № 6

1. $x^4 y'' + x^3 y' = 1$
2. $y' + y\, tgx = \frac{1}{\cos x}$ $\qquad y(0) = 0$
3. $ye^{2x}dx - (1 + e^{2x})dy = 0$
4. $4y'' y^3 = -1$
5. $y'' + y' = 2x^2 e^x$
6. $2(y' + y) = xy^2$ $\qquad y(0) = 2$
7. $y' = \frac{x + y}{x - y}$
8. $y'' + \pi^2 y = \frac{\pi^2}{\sin \pi x}$
9. $y'' + 6y' + 13y = e^{-3x} \cos x$

Вариант № 2

1. $y'' + 9y = \frac{9}{\sin^3 x}$ $\qquad y\left(\frac{\pi}{6}\right) = 4$, $\quad y'\left(\frac{\pi}{6}\right) = \frac{3\pi}{2}$
2. $y' - 2xy = 3x^3 y^2$
3. $xy' - 2y = 2x^4 y$
4. $(1 - x^2)y'' + xy' - 2 = 0$
5. $y' = \frac{2y}{x + 1} + e^x (x + 1)^2$
6. $y^2 y' = x^2 + \frac{y^3}{x}$
7. $y'' + y = 2\sin x$
8. $y'' y^3 + 25 = 0$ $\qquad y(2) = -5$, $\quad y'(2) = -1$
9. $y'' - 5y' + 6y = 6x^2 + 2x - 5$